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ELECOURSE 2026 Spring Project: Development of an End-to-End ECG/PPG-Based Cuffless Blood Pressure Estimation System

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Abstract

Continuous blood pressure monitoring is essential for the prevention and management of cardiovascular diseases. Conventional cuff-based blood pressure measurement methods are unsuitable for continuous monitoring due to discomfort and interruption of daily activities. In this study, an end-to-end cuffless blood pressure estimation pipeline based on electrocardiography (ECG) and photoplethysmography (PPG) signals was developed. The proposed system integrates hardware design, biosignal acquisition, signal processing, feature extraction, machine learning-based estimation, and experimental validation. ECG signals were acquired using an AD8232 module, while PPG signals were collected using a MAX30102 sensor. Pulse Arrival Time (PAT) and Pulse Transit Time (PTT) were extracted from synchronized ECG and PPG signals and used together with demographic features including age, sex, height, and weight. Data were collected from 20 healthy participants, resulting in 84 measurement samples. Three regression models—Support Vector Regression (SVR), Random Forest (RF), and Hybrid LASSO-XGBoost—were evaluated. Experimental results demonstrated that the Hybrid LASSO-XGBoost model achieved the best overall performance, with mean absolute errors of 6.90 mmHg and 6.74 mmHg for systolic and diastolic blood pressure estimation, respectively. The results demonstrate the feasibility of ECG/PPG-based cuffless blood pressure estimation and provide insights for future wearable healthcare applications.

Keywords - Cuffless Blood Pressure Estimation, ECG, PPG, Pulse Transit Time, Pulse Arrival Time, Machine Learning, Wearable Healthcare

1. Introduction

Hypertension is a major risk factor for cardiovascular disease, stroke, heart failure, and kidney failure, and is one of the most prevalent chronic diseases worldwide. Blood pressure is a key vital sign that reflects an individual's cardiovascular health status, and continuous monitoring of blood pressure plays an important role not only in the early detection and prevention of diseases but also in the long-term management of patient health. In particular, with the rapid growth of the aging population and lifestyle changes, the number of individuals with hypertension has been increasing, leading to growing interest in technologies that enable continuous blood pressure monitoring during daily activities.

Currently, the most widely used method for blood pressure measurement is the cuff-based oscillometric method. However, this approach requires periodic compression of the arm, which restricts user movement and causes discomfort during repeated measurements over extended periods. Furthermore, it only provides blood pressure values at discrete measurement times, making it difficult to observe continuous physiological changes such as blood pressure variations during sleep, exercise, or stressful situations. For these reasons, cuffless blood pressure estimation technologies that can monitor blood pressure without interfering with daily activities have recently attracted significant attention as next-generation digital healthcare solutions.

With the advancement of wearable healthcare technologies, numerous non-invasive blood pressure estimation methods utilizing various biosignals have been investigated. Among them, approaches that simultaneously employ Electrocardiography (ECG) and Photoplethysmography (PPG) are particularly attractive because they are relatively easy to implement and well suited for wearable devices. ECG measures the electrical activity of the heart, whereas PPG measures peripheral blood volume changes using optical sensors. The time difference between these signals, known as Pulse Arrival Time (PAT) or Pulse Transit Time (PTT), has been shown to correlate strongly with blood pressure and can therefore be used for indirect blood pressure estimation. Recent studies have further improved estimation accuracy by integrating these physiological features with machine learning and deep learning techniques.

This project was conducted as part of the regular project activities of ELECOURSE, an undergraduate electrical and electronics engineering consortium club. ELECOURSE consists of students from various universities who collaborate on engineering projects spanning hardware design, embedded systems, signal processing, artificial intelligence, and robotics. During this semester, the development of a cuffless blood pressure estimation system integrating wearable healthcare technology and artificial intelligence was selected as the project topic. The project was carried out through collaboration among the Hardware (HW), Signal Processing (SP), Machine Learning (ML), and Validation (VL) teams.

In this study, an STM32-based biosignal acquisition system capable of simultaneously collecting ECG and PPG signals was designed and implemented. The acquired signals were processed through filtering, R-peak detection, PPG foot point detection, and PAT/PTT calculation. In addition, demographic information including age, gender, height, and weight was incorporated into the blood pressure prediction model. Data were collected from 20 subjects through repeated measurements, resulting in a total of 84 measurement samples.

To estimate blood pressure, Support Vector Regression (SVR), Random Forest (RF), and a Hybrid LASSO-XGBoost model were implemented and compared. Furthermore, Signal Quality Index (SQI), data validation procedures, and subject-independent validation techniques were employed to evaluate model generalizability and physiological validity. Experimental results demonstrated that the Hybrid LASSO-XGBoost model achieved the best performance, with mean absolute errors (MAE) of 6.90 mmHg and 6.74 mmHg for systolic blood pressure (SBP) and diastolic blood pressure (DBP), respectively.

The primary objective of this study is to verify the feasibility of an ECG- and PPG-based cuffless blood pressure estimation system under limited data conditions and to establish a complete pipeline encompassing hardware design, signal processing, machine learning-based prediction, and validation. In addition, this project aims to provide foundational research data for the future development of wearable healthcare devices and continuous blood pressure monitoring systems.

2. Materials and Methods

In this study, a cuffless blood pressure estimation system based on Electrocardiography (ECG) and Photoplethysmography (PPG) signals was developed and evaluated. The proposed system consists of biosignal acquisition hardware, signal preprocessing, feature extraction, machine learning-based blood pressure prediction, and performance validation modules.

First, ECG and PPG signals were synchronously acquired from each subject using dedicated sensors. The acquired signals were then processed through filtering and noise reduction procedures. Subsequently, R-peaks from ECG signals and characteristic points from PPG signals were detected to calculate Pulse Arrival Time (PAT) and Pulse Transit Time (PTT). Demographic information, including age, gender, height, and weight, was also incorporated as additional input features.

The extracted features were used as inputs to Support Vector Regression (SVR), Random Forest (RF), and Hybrid LASSO-XGBoost models, and the predictive performances of these models were comparatively evaluated. Experiments were conducted on 20 healthy adult subjects, with systolic blood pressure (SBP) and diastolic blood pressure (DBP) serving as prediction targets. This chapter first describes the hardware configuration of the biosignal measurement system, followed by the data acquisition procedure, signal processing and feature extraction methods, machine learning model architectures, and performance evaluation methodologies.

2.1. Hardware System Design

In this study, a biosignal acquisition system capable of simultaneously collecting Electrocardiography (ECG) and Photoplethysmography (PPG) signals was developed for cuffless blood pressure estimation. An AD8232 module was employed for ECG signal acquisition, while a MAX30102 sensor was used for PPG signal measurement. An STM32 microcontroller was utilized for integrated signal management and data transmission.

The AD8232 is an analog front-end (AFE) module capable of single-lead ECG measurement. It incorporates built-in amplification and filtering circuits, making it suitable for stable acquisition of electrocardiographic signals. The MAX30102 is an optical sensor that measures blood volume changes using red and infrared LEDs and is widely used in wearable biosignal monitoring systems. The STM32 microcontroller provides multiple ADC and I²C interfaces, enabling simultaneous acquisition and processing of ECG and PPG signals.

The overall system architecture was designed such that signals acquired from the AD8232 ECG sensor and the MAX30102 PPG sensor were collected by the STM32 and subsequently transmitted to a PC via USB serial communication. Since the ECG signal is output in analog form, it was converted into digital data using the STM32's Analog-to-Digital Converter (ADC). In contrast, the PPG signal was processed as digital data within the MAX30102 and acquired directly through the I²C communication interface. Additionally, the Lead-off Detection output of the AD8232 was connected to the STM32 GPIO pins, allowing real-time monitoring of electrode contact status.

The MAX30102 was interfaced with the STM32 through the I²C3 peripheral. The acquired PPG data were synchronized and stored together with the ECG data within the STM32 before being transmitted to a PC via USB serial communication. This integrated architecture enabled efficient acquisition of heterogeneous biosignals on a single platform.

The hardware configuration employed in this study allows simultaneous measurement of ECG and PPG signals and provides the temporal synchronization information required for calculating Pulse Arrival Time (PAT) and Pulse Transit Time (PTT). Therefore, the proposed system can serve as a biosignal acquisition platform for non-invasive continuous blood pressure estimation.

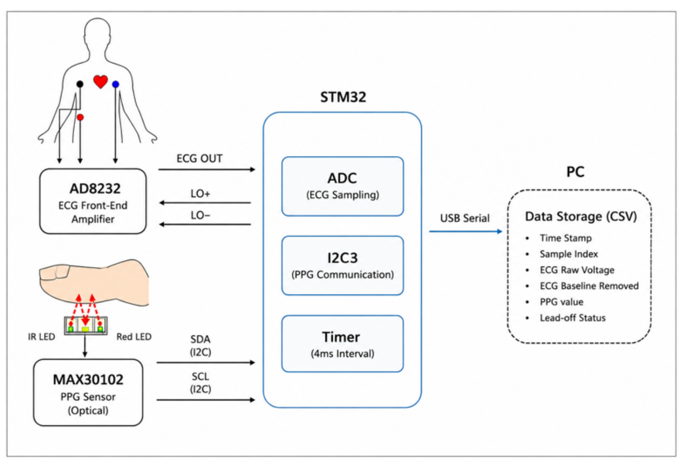


Fig.1. Overall architecture of the ECG/PPG acquisition system

2.2. Signal Aquisition

ECG and PPG signals were acquired simultaneously using an STM32-based embedded data acquisition system. For ECG measurement, an AD8232 module was employed, and electrodes were attached using a standard

Fig. 2. Signal Processing and Feature Extraction Pipeline

three-electrode configuration on the left chest, right chest, and right lower torso. PPG signals were measured at the fingertip using a MAX30102 sensor.

Biosignals were collected at intervals of approximately 4 ms based on the STM32 internal timer, resulting in an effective sampling frequency of approximately 250 Hz. The acquired data included timestamps, sample indices, raw ECG signals, baseline-drift-corrected ECG signals, PPG signals, and lead-off status information. Each sample was assigned a unique sample index to facilitate detection of data loss, while timestamp information was used to preserve the temporal relationships among measurements.

The collected data were transmitted to a PC in real time via USB serial communication and stored in CSV format using a Python-based data logging program. In addition, the Lead-off Detection function of the AD8232 was utilized to monitor electrode contact status, ensuring that only reliable data were used for subsequent analysis.

To obtain reference blood pressure values for model training and evaluation, systolic blood pressure (SBP) and diastolic blood pressure (DBP) were measured simultaneously using a commercial automatic blood pressure monitor during ECG and PPG acquisition. Subject information, including age, gender, height, weight, caffeine intake status, and exercise status, was also recorded and incorporated into the dataset.

All measurements were conducted under resting conditions while subjects were seated comfortably. To minimize motion artifacts and measurement noise, unnecessary body movements were restricted throughout the data acquisition process.

2.3. Signal Processing

The acquired ECG and PPG signals underwent preprocessing prior to feature extraction. First, data continuity was verified using the sample indices to ensure data integrity. Segments containing data losses exceeding a predefined threshold were separated and processed independently.

For ECG signal processing, DC offset removal, power-line noise suppression, and band-pass filtering were performed. After removing the DC offset, a 60 Hz notch filter was applied to eliminate power-line interference. Subsequently, a 0.5–40 Hz Butterworth band-pass filter was used to suppress baseline wander and high-frequency noise. Zero-phase filtering was employed throughout the filtering process to minimize phase distortion.

For PPG signal preprocessing, a 0.5–7 Hz Butterworth band-pass filter was applied to facilitate pulse waveform feature extraction. This filtering process simultaneously removed low-frequency components caused by respiration and motion artifacts as well as high-frequency noise, thereby enabling more accurate detection of the PPG waveform upstroke and foot point.

2.4. Feature Extraction

The primary features used for blood pressure estimation were the ECG R-peaks and the PPG foot points. Pulse Arrival Time (PAT) and Pulse Transit Time (PTT) were subsequently calculated using these extracted fiducial points. The overall feature extraction process consisted of ECG R-peak detection, PPG foot point detection, PAT/PTT calculation, and data validation.

2.4.1 ECG R-Peak Detection

R-peak detection was performed using the filtered ECG signal. In this study, a method similar to the Pan–Tompkins algorithm was adopted to enhance the QRS complex and detect R-peaks.

First, the ECG signal was differentiated to emphasize regions with rapid changes, such as QRS complexes. The differentiated signal was then squared so that all values became positive and larger signal variations were further amplified. Subsequently, moving window integration was applied to

generate a QRS energy signal. The moving integration window length was set to 0.12 s.
A dynamic threshold based on the mean and standard deviation of the QRS energy signal was then established. Peaks exceeding the threshold were selected as R-peak candidates.

However, peaks detected from the QRS energy signal may not exactly correspond to the true R-peak locations in the ECG waveform. To refine the detection, the local maximum of the filtered ECG signal was searched within ±0.08 s around each candidate peak. The identified local maximum was designated as the final R-peak location. A signal segment was considered valid only when the number of detected R-peaks exceeded a predefined minimum threshold.

2.4.2 PPG Foot Point Detection

PPG foot point detection was performed using ECG R-peaks as temporal references rather than through an independent peak detection algorithm. This approach exploits the physiological timing relationship between ECG and PPG signals and reduces the likelihood of false detections.
First, the filtered PPG signal was differentiated to obtain the Velocity Photoplethysmogram (VPG). The VPG was then differentiated again to generate the Acceleration Photoplethysmogram (APG). In this study, VPG information was primarily used for foot point detection.

For each ECG R-peak, a search window between 0.10 s and 0.45 s after the R-peak was defined as the candidate region for PPG foot point detection. This range reflects the typical physiological delay between cardiac electrical activation and pulse wave arrival at peripheral vessels.

Within the search window, the point corresponding to the maximum VPG value was defined as the Maximum Slope Point. A local minimum of the PPG waveform was then identified within 0.06 s preceding the Maximum Slope Point. Finally, the PPG foot point was determined using the Tangent Intersection Method. Specifically, the foot point was defined as the intersection between the tangent line at the Maximum Slope Point and the baseline amplitude line passing through the local minimum. Compared with simple minimum-value detection methods, this approach more effectively captures the morphology of the PPG upstroke.

2.4.3 PAT/PTT Calculation

PAT and PTT were calculated after matching ECG R-peaks and PPG foot points corresponding to the same cardiac cycle.

First, the RR interval was calculated as the time difference between consecutive R-peaks. PAT was defined as the time difference between the ECG R-peak and the corresponding PPG foot point:

$$PAT = t_{foot} - t_R$$

where (t_R) denotes the occurrence time of the ECG R-peak and (t_{foot}) denotes the occurrence time of the corresponding PPG foot point.

PTT was defined as PAT minus the Pre-Ejection Period (PEP):

$$PTT = PAT - PEP$$

where (t_R) denotes the occurrence time of the ECG R-peak and (t_{foot}) denotes the occurrence time of the corresponding PPG foot point.

PTT was defined as PAT minus the Pre-Ejection Period (PEP):

$$PTT_{ms} = PAT_{ms} - 70$$

The calculated results were stored in a beat-level feature table containing the following variables:

2.4.4 Data Validation

To ensure the reliability of the calculated PAT and PTT values, physiological validity checks were performed through a multi-stage data validation procedure.

In the first stage, absolute range validation was applied. Beats with PEP, PAT, or PTT values outside predefined physiological ranges were excluded. The ranges used in this study were:

- PEP: 20–180 ms
- PAT: 150–450 ms
- PTT: 80–380 ms

In the second stage, median-based outlier detection was performed. PAT and PTT values that deviated excessively from the segment median were classified as outliers. For PAT, the criterion was ±15% of the median or a minimum threshold of 30 ms. For PTT, the criterion was ±20% of the median or a minimum threshold of 30 ms.

In the third stage, beat-to-beat variation analysis was conducted. Beats exhibiting PAT or PTT changes greater than 55 ms compared to adjacent beats were regarded as abnormal detections and removed.

Finally, beats with RR intervals outside ±30% of the segment median were excluded. Only beats satisfying all validation criteria were retained as final feature extraction results.

2.5. Machine Learning Models

The ultimate objective of this study was to predict systolic blood pressure (SBP) and diastolic blood pressure (DBP) using biosignal-derived features and demographic information extracted from ECG and PPG signals.

Age, Gender, Height, Weight, PAT, and PTT were used as input features. SBP and DBP prediction were formulated as separate regression problems, and machine learning models were trained accordingly.

2.5.1 Dataset Construction

During the initial development stage, an artificially generated nonlinear-function dataset was utilized to verify the functionality of the machine learning pipeline.

Subsequently, several publicly available datasets were investigated. However, no dataset containing all variables required for this study (SBP, DBP, PAT, PTT, Height, Weight, and Gender) was identified.

Therefore, the research team constructed a proprietary dataset. ECG and PPG signals were collected from 20 healthy adult subjects. Repeated measurements were performed for each subject, resulting in a total of 84 samples. Each sample contained information on SBP, DBP, Age, Gender, Height, Weight, PAT, and PTT.

To mitigate the limitations of the relatively small dataset, data augmentation techniques were applied in selected experiments. Augmented samples were generated while preserving the statistical characteristics of the original dataset. The effectiveness of data augmentation was evaluated by comparing model performance before and after augmentation.

2.5.2 Support Vector Regression (SVR)

Support Vector Regression (SVR) is a regression model derived from Support Vector Machines (SVM) that learns the relationship between input variables and target variables based on the maximum-margin principle.

In this study, a Radial Basis Function (RBF) kernel was employed to model the nonlinear relationship between PAT/PTT and blood pressure. SVR is known for its stable performance with relatively small training datasets and its low susceptibility to overfitting. Furthermore, its ability to effectively capture complex nonlinear relationships between physiological signals and blood pressure made it an appropriate baseline model for this study.

The primary hyperparameters are listed below:

- **C**: Penalty coefficient controlling prediction errors
- **ε (epsilon)**: Width of the insensitive loss region
- **γ (gamma)**: Influence range of the RBF kernel
- **cv**: Number of cross-validation folds
- **n_iter**: Number of RandomizedSearchCV iterations

The optimal hyperparameters were determined using RandomizedSearchCV.

2.5.3 Random Forest

Random Forest (RF) is an ensemble-based regression model that performs prediction by aggregating multiple decision trees. Each tree is trained independently using randomly selected subsets of data samples and input features, and the final prediction is obtained by averaging the outputs of all trees. RF is capable of effectively modeling nonlinear relationships among variables and provides feature importance measures, thereby enhancing model interpretability. For this reason, RF was employed as a comparative model to analyze the influence of physiological features such as PAT and PTT on blood pressure prediction.

The primary hyperparameters of the RF model are as follows:

- **n_estimators**: Number of decision trees
- **max_depth**: Maximum depth of each tree
- **min_samples_split**: Minimum number of samples required to split a node
- **random_state**: Random seed
- **n_trials**: Number of optimization iterations in the Optuna framework

Hyperparameter optimization was performed using the Optuna optimization framework.

2.5.4 Hybrid LASSO-XGBoost

To leverage the strengths of both linear and nonlinear models, a Hybrid LASSO-XGBoost model was proposed in this study. LASSO is a linear regression model that incorporates L1 regularization, which drives the coefficients of irrelevant features toward zero and thereby performs implicit feature selection. XGBoost is a gradient-boosting-based tree ensemble model that effectively captures complex nonlinear relationships.

The proposed Hybrid model consists of two stages. First, the LASSO model learns the linear component of blood pressure from the input features. Subsequently, XGBoost is trained on the residual errors that are not explained by LASSO, generating the final prediction. This architecture was designed to simultaneously alleviate the underfitting tendency of linear models and the overfitting tendency of nonlinear models.

The primary hyperparameters of the LASSO model include the regularization strength (**α**), maximum number of iterations (**max_iter**), and convergence tolerance (**tolerance**). For the XGBoost model, the primary hyperparameters include **n_estimators**, **learning_rate**, **max_depth**, **subsample**, **colsample_bytree**, **reg_alpha**, and **reg_lambda**. The optimal hyperparameters for both models were determined using cross-validation-based RandomizedSearchCV.

2.5.5 Training and Evaluation Procedure

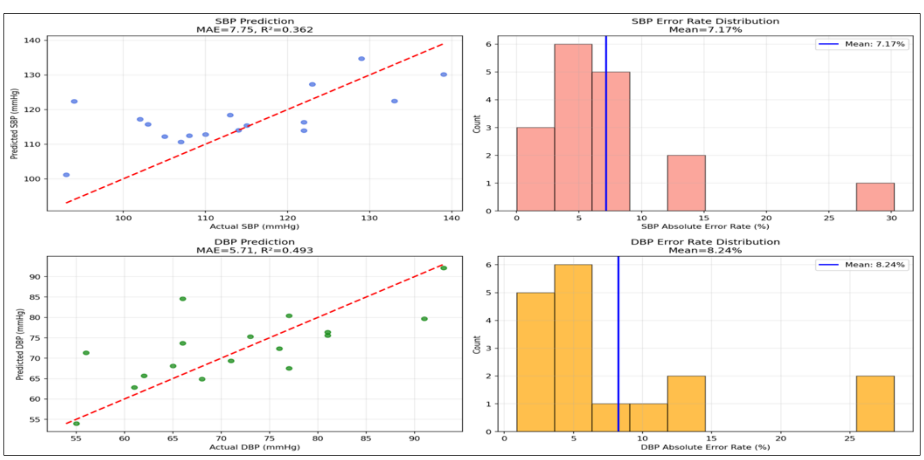
Prior to model training, the dataset was cleaned by removing missing values and outliers. The resulting dataset was then divided into training and testing subsets. Depending on the model and experimental setting, Internal Split, Leave-One-Group-Out (LOGO) cross-validation, and External Test validation strategies were employed.

For the SVR and Hybrid models, standard scaling was applied to mitigate the influence of differences in feature magnitudes and units. In contrast, the Random Forest model was trained without feature scaling due to the inherent properties of tree-based algorithms.

During model development, hyperparameter optimization was performed using Grid Search, Randomized Search, and Optuna-based optimization methods. After training, each model was used to predict systolic blood pressure (SBP) and diastolic blood pressure (DBP) on the validation dataset. Model performance was evaluated using Mean Absolute Error (MAE), Mean Error (ME), Mean Squared Error (MSE), and the coefficient of determination (R^2).

To further investigate model behavior and interpretability, several visualization and analysis techniques were employed, including Predicted-versus-Actual plots, error distribution analysis, correlation heatmaps, and feature importance analysis.

Fig.3. SVR Model Prediction Result



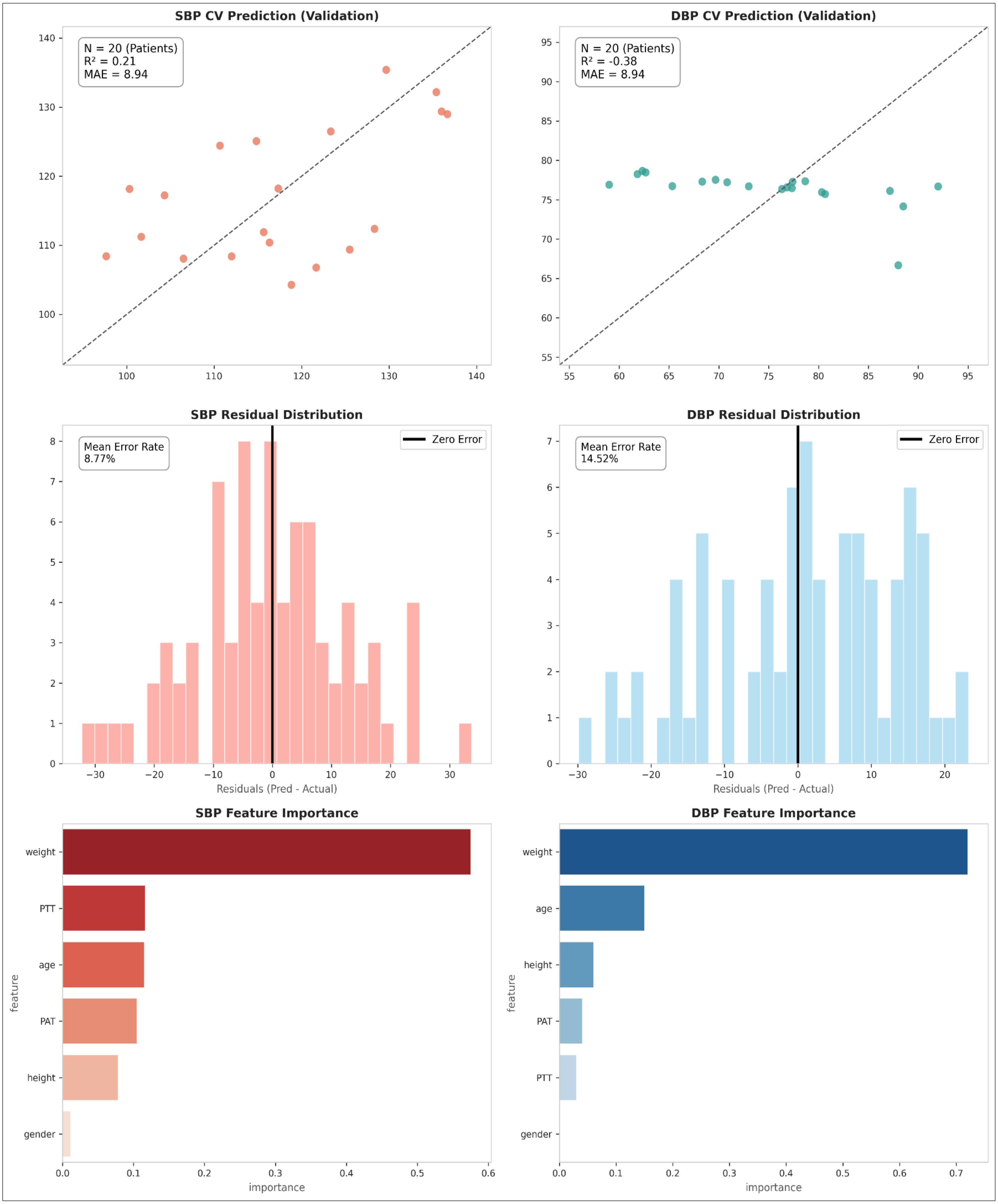


Fig.4. Random Forest Model Prediction Result

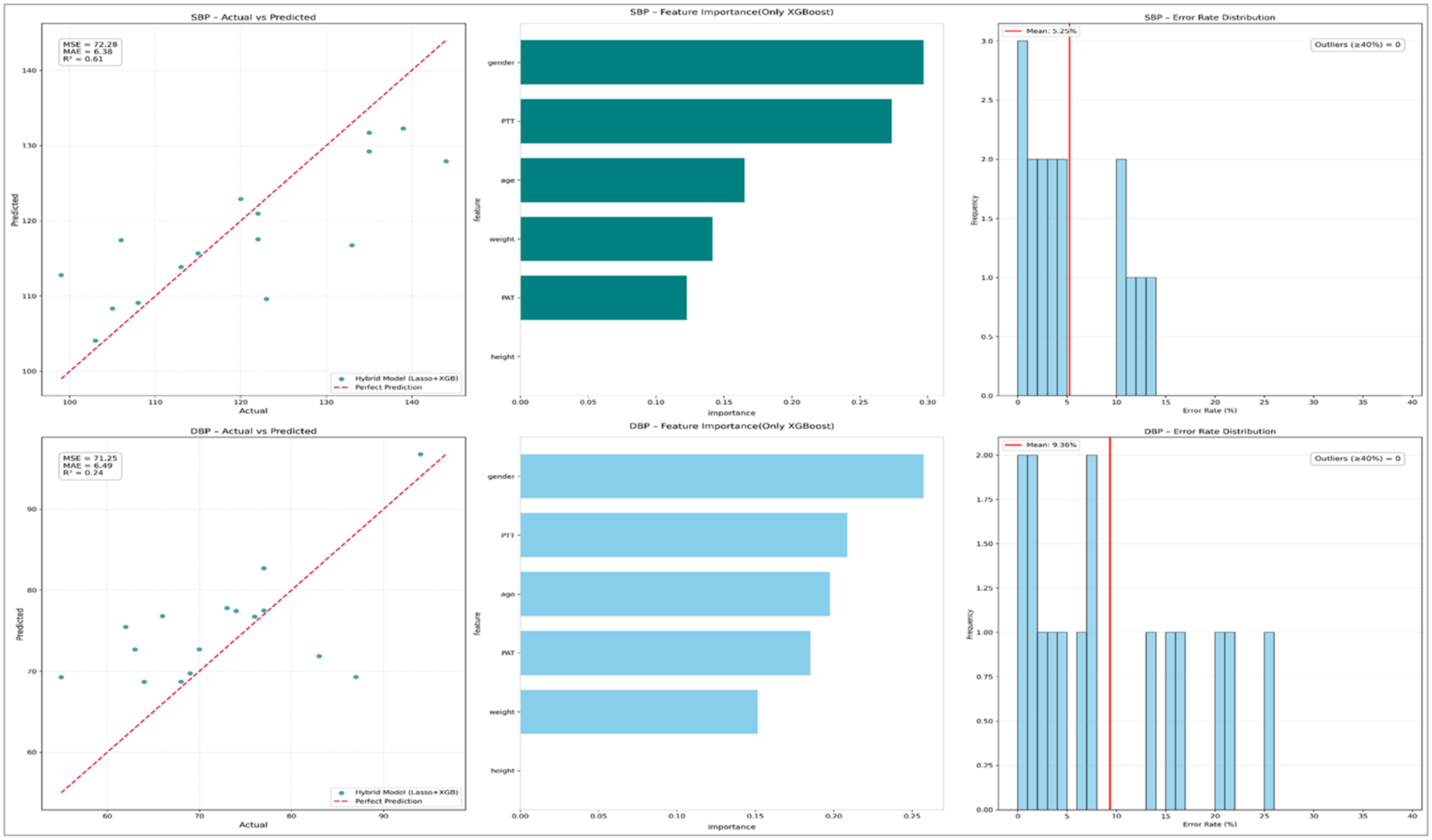


Fig.5. Hybrid Lasso + XGBoost Model Prediction Result

2.6. Experimental Protocol and Validation Criteria

To ensure the reliability of the proposed ECG- and PPG-based non-invasive blood pressure estimation system, a comprehensive experimental framework encompassing measurement conditions, subject selection, signal quality management, and performance evaluation criteria was established. In particular, a standardized measurement protocol was adopted to minimize the influence of physiological and environmental factors on blood pressure and biosignal acquisition.

2.6.1 Subject Selection and Measurement Conditions

Healthy adult volunteers with no history of cardiovascular disease or hypertension were recruited for this study. To reduce variability in baseline hemodynamic conditions among subjects, the age range was restricted to 20–25 years.

To control external factors that may affect blood pressure, all participants were instructed to refrain from caffeine consumption, smoking, and vigorous physical activity for at least 30 minutes prior to measurement. In addition, participants were asked to empty their bladder before the experiment to eliminate potential sympathetic nervous system stimulation caused by bladder distension.

All measurements were conducted in an indoor environment. Subjects remained seated comfortably in a chair with back support throughout the measurement procedure. Leg crossing, which may induce hemodynamic changes, was prohibited, and the arm equipped with sensors was maintained at heart level. Data acquisition began after a minimum resting period of five minutes, and at least three repeated measurements were obtained from each subject.

2.6.2 Measurement Environment Control

To minimize external noise during biosignal acquisition, participants were instructed to avoid conversation, smartphone use, and unnecessary body movements throughout the measurement process. In addition, external light sources were minimized to reduce optical interference with the PPG sensor. Sensor connections and signal transmission conditions were continuously monitored to mitigate electromagnetic interference and ensure data quality.

2.6.3 Signal Quality Assessment and Data Rejection

A Signal Quality Index (SQI)-based validation procedure was applied to quantitatively assess the quality of the collected biosignals.

For PPG signals, quality assessment was performed using template matching, skewness, kurtosis, and zero-crossing rate metrics. For ECG signals, reliability was evaluated by comparing the consistency of multiple R-peak detection results.

Data satisfying any of the following conditions were excluded from analysis:

1. Signal clipping resulting in prolonged waveform saturation.
2. SQI scores below predefined quality thresholds.
3. Heart rate values outside physiologically plausible ranges.
4. Severe waveform distortion caused by motion artifacts, poor electrode contact, or external interference.

Only data that satisfied all quality requirements were retained for machine learning model training and evaluation.

2.6.4 Physiological Interpretation of PTT

This study is based on the assumption that the relationship between Pulse Transit Time (PTT) and blood pressure is governed by arterial elasticity and pulse wave propagation velocity. This relationship can be explained by the Moens–Korteweg equation, which suggests that increases in blood pressure lead to increased arterial stiffness, resulting in faster pulse wave propagation and consequently shorter PTT values.

However, under real-world conditions, this relationship may vary due to factors such as arterial diameter, vascular elasticity, autonomic nervous system activity, and individual physiological characteristics. Therefore, in addition to PTT, demographic variables including age, gender, height, and weight were incorporated into the blood pressure estimation models.

2.6.5 Performance Evaluation Criteria

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Error (ME), and the coefficient of determination (R^2). The detailed performance results of the evaluated models are presented in Table 1.

To facilitate clinical interpretation, the results were assessed with reference to the international standard for non-invasive blood pressure monitoring devices, ISO 81060-2. According to this standard, a device should achieve a mean error of no more than 5 mmHg and a standard deviation of error below 8 mmHg. Consequently, both absolute error metrics (mmHg) and relative performance indicators were considered when evaluating the clinical applicability of the proposed system.

Furthermore, subject-independent validation was performed using a Leave-One-Subject-Out (LOSO) cross-validation strategy. This approach prevents data leakage by ensuring that data from the same subject are never used simultaneously for training and evaluation, thereby providing a more realistic assessment of model generalization to previously unseen users.

Table 1

Algorithm Validation Results

Dataset Configuration (Data Augmentation)	**Algorithm**	**Validation Team Assessment (Metrics and Observations)**	**Decision**
Dataset(3)_60 (Without Augmentation)	**SVR**	Failed to learn physiological signal patterns due to insufficient data. Loss of blood pressure trend tracking was observed.	**Reject**
Dataset(3)_110 (Without Augmentation)	**Random Forest**	Feature importance analysis indicated severe overfitting. Predictions were highly dependent on demographic variables rather than physiological features.	**Reject**
Dataset(3)_100 (With Augmentation)	**LASSO + XGBoost**	Successfully captured the increasing trend of actual blood pressure values. Prediction errors for both SBP and DBP were concentrated within the 1–5% range.	**Pass**

3. Results

This section presents the implementation results of the ECG- and PPG-based cuffless blood pressure estimation system, including signal acquisition, data processing, and machine learning-based blood pressure prediction performance. The results are organized into signal acquisition, feature extraction, blood pressure estimation performance, and model comparison.

3.1. Signal Acquisition Results

In this study, a biosignal acquisition system based on the STM32 microcontroller was developed to simultaneously collect ECG and PPG signals. The system employed an AD8232 ECG sensor and a MAX30102 PPG sensor, acquiring data at a sampling frequency of approximately 250 Hz. Timestamp and Sample Index information were recorded for every sample to facilitate subsequent signal processing and synchronization.

During the initial development stage, several technical issues were encountered, including ECG baseline drift, PPG sensor communication errors, and instability of the I^2C interface. In particular, the MAX30102 sensor exhibited unreliable operation on the STM32 I^2C1 interface. This issue was resolved by migrating the sensor communication to the I^2C3 channel, after which stable data acquisition was achieved. Furthermore, a Lead-off Detection function was implemented to monitor ECG electrode contact status in real time.

The finalized system was capable of simultaneously storing both the raw ECG signal and the baseline-corrected ECG signal. Stable ECG and PPG acquisition was achieved during long-term measurements. Collected data were transmitted to a PC in real time via USB serial communication and stored in CSV format for subsequent signal processing and machine learning model training.

The following major software modules were developed throughout the project:

- ECG/PPG Filtering Module
- PAT/PTT Extraction Module
- Signal Validation Module
- Feature Extraction Module
- Blood Pressure Estimation Module

3.2 PAT/PTT Extraction Results

The proposed signal processing pipeline was applied to the acquired ECG and PPG signals to detect ECG R-peaks and PPG foot points.

For ECG processing, a 60 Hz notch filter and a 0.5–40 Hz band-pass filter were applied, followed by R-peak detection using a Pan–Tompkins-based algorithm. For PPG processing, a 0.5–7 Hz band-pass filter was applied, and foot points were detected using a VPG-based foot detection algorithm.

The detected ECG R-peaks and PPG foot points were matched on a beat-by-beat basis to calculate Pulse Arrival Time (PAT) and Pulse Transit Time (PTT). Subsequently, a data validation procedure was performed to eliminate physiologically implausible measurements. The validation process included range checks for PAT, PTT, and RR intervals, median-based outlier removal, and beat-to-beat variation analysis.

The following features were ultimately extracted for each heartbeat:

- PAT (Pulse Arrival Time)
- PTT (Pulse Transit Time)
- RR Interval
- Age
- Gender
- Height
- Weight

These features were subsequently used as input variables for the blood pressure prediction models.

A total of 84 valid measurement samples were collected from 20 participants. All samples successfully passed the data validation procedure and were utilized for machine learning model training and evaluation.

3.3 Blood Pressure Estimation Results

Three machine learning models—Support Vector Regression (SVR), Random Forest (RF), and Hybrid LASSO-XGBoost—were trained for blood pressure estimation.

The input features consisted of Age, Gender, Height, Weight, PAT, and PTT, while the target variables were systolic blood pressure (SBP) and diastolic blood pressure (DBP).

Model performance was evaluated using Mean Absolute Error (MAE), Mean Error (%), Mean Squared Error (MSE), and the coefficient of determination (R^2).

3.3.1 SVR Performance

The performance results of the SVR model are summarized in Table 2.

Table 2

Metric	SBP	DBP
MAE (mmHg)	7.75	5.71
R^2	0.362	0.493
Mean Error (%)	7.17	8.24

The SVR model demonstrated relatively strong performance in DBP prediction and achieved positive R^2 values for both SBP and DBP, indicating its ability to capture the general trend of blood pressure variations to a certain extent. However, when Data Augmentation was applied, model performance deteriorated, as shown in Table 3.

Table 3

Metric	SBP	DBP
MAE (mmHg)	15.10	11.91
R^2	-2.262	-2.221
Mean Error (%)	12.04	16.98

3.3.2 Random Forest Performance

The Random Forest model was evaluated using Leave-One-Group-Out (LOGO) cross-validation. The corresponding performance results are presented in Table 4.

Table 4

Metric	SBP	DBP
MAE (mmHg)	8.94	8.94
R^2	0.21	-0.38
Mean Error (%)	8.77	14.52

The Random Forest model exhibited limited predictive capability for SBP and produced negative R^2 values for DBP, suggesting insufficient generalization performance. Feature importance analysis revealed that Weight and Age contributed most significantly to the model predictions, whereas the contributions of PAT and PTT were relatively limited.

3.3.3 Hybrid LASSO-XGBoost Performance

The Hybrid LASSO-XGBoost model was implemented using a two-stage architecture in which LASSO first learned the linear relationship between the input features and blood pressure, and XGBoost subsequently modeled the residual errors that remained unexplained by the linear model. Among the three evaluated models, the Hybrid model achieved the best overall performance. The detailed performance results are presented in Table 5.

Table 5

Metric	SBP	DBP
MAE (mmHg)	6.90	6.74
Mean Error (%)	~ 5%	~ 5% 수

The Hybrid model achieved the lowest MAE values for both SBP and DBP and demonstrated the most stable ability to track actual blood pressure trends among all evaluated models.

4. Discussion

In this study, a cuffless blood pressure estimation system based on ECG and PPG signals was developed, integrating the entire pipeline from biosignal acquisition and feature extraction to machine learning-based blood pressure prediction and performance validation. A total of 84 measurement samples were collected from 20 subjects, and systolic blood pressure (SBP) and diastolic blood pressure (DBP) were estimated using PAT, PTT, and demographic information. Experimental results demonstrated that the Hybrid LASSO-XGBoost model achieved the best overall performance, while the SVR and Random Forest models exhibited relatively lower generalization capability.

4.1. Effectiveness of PAT/PTT-Based Blood Pressure Estimation

PAT and PTT, utilized in this study, are representative blood pressure-related features that describe the temporal relationship between ECG and PPG signals. It is well established that increased blood pressure is associated with increased arterial stiffness, resulting in faster pulse wave propagation and consequently shorter PTT values. Based on this physiological relationship, PAT and PTT were selected as the primary features for blood pressure estimation.

The experimental results demonstrated that all evaluated models achieved a certain level of blood pressure prediction performance, supporting the feasibility of non-invasive blood pressure estimation using ECG and PPG signals. However, PAT and PTT alone were insufficient to fully explain an individual's absolute blood pressure level. In particular, substantial performance degradation was observed when predicting blood pressure for previously unseen subjects. This finding suggests that blood pressure is influenced not only by pulse wave propagation time but also by various physiological factors, including arterial elasticity, peripheral vascular resistance, and autonomic nervous system activity.

4.2. Machine Learning Model Comparison

Three regression models were evaluated in this study: Support Vector Regression (SVR), Random Forest (RF), and Hybrid LASSO-XGBoost.

SVR demonstrated relatively stable performance despite the limited dataset size and achieved the lowest MAE for DBP prediction. However, its performance deteriorated when data augmentation was applied. This behavior may be attributed to the sensitivity of RBF-kernel-based SVR to the geometric structure of the feature space. Synthetic samples that did not accurately reflect physiological distributions likely introduced additional noise and negatively affected model performance.

The Random Forest model effectively captured nonlinear relationships but exhibited excessive reliance on demographic variables such as body weight and age, according to feature importance analysis. In contrast, the importance of physiologically meaningful variables such as PAT and PTT was relatively low. This finding suggests that the model tended to memorize subject-specific characteristics rather than learn the underlying physiological mechanisms of blood pressure regulation, representing a typical case of overfitting in a small-data environment.

Among all evaluated models, the Hybrid LASSO-XGBoost model achieved the best performance. LASSO first learns linear relationships and removes irrelevant variables, while XGBoost subsequently learns the residual errors and captures nonlinear relationships. This hybrid architecture effectively mitigated both overfitting and underfitting issues commonly encountered in small datasets. The Hybrid model achieved the lowest mean absolute errors for both SBP and DBP and demonstrated the most stable tracking of blood pressure trends.

4.3. Validation Strategy and Clinical Interpretation

Beyond conventional prediction accuracy assessment, this study established a validation framework that considered physiological plausibility and potential clinical applicability. During data collection, factors known to influence blood pressure, such as caffeine intake, physical activity, and posture changes, were controlled. In addition, Signal Quality Index (SQI) evaluation and predefined data rejection criteria were applied to ensure that only reliable data were included in the analysis.

For model validation, Leave-One-Group-Out (LOGO) cross-validation was employed to evaluate performance on subjects not included in the training set. This approach minimizes data leakage that may occur in conventional random train-test splits and provides a more rigorous assessment of model generalization under realistic wearable monitoring scenarios.

From a clinical perspective, the results were interpreted with reference to the AAMI/ESH/ISO 81060-2 guidelines. Although the Hybrid model achieved relatively low MAE values, several samples exhibited large prediction errors, indicating that the proposed system has not yet reached the performance level required for clinical-grade blood pressure monitoring devices. In particular, the outliers observed in DBP prediction remain an important challenge for future practical deployment.

4.4. Limitations and Future Work

This study has several limitations.

First, the dataset size was limited. The dataset consisted of only 20 subjects and 84 measurement samples, which is insufficient for establishing robust model generalization. Furthermore, the LOGO validation strategy further reduced the amount of training data available in each fold.

Second, the reference blood pressure values were obtained using a commercial blood pressure monitor rather than through simultaneous invasive measurements. Consequently, temporal discrepancies between biosignal acquisition and reference blood pressure measurements may have introduced additional variability into the ground-truth labels.

Third, the feature set was limited to PAT, PTT, and basic demographic variables. Incorporating additional physiological features such as arterial stiffness indicators, pulse waveform morphology, and heart rate variability (HRV) may further improve prediction performance.

Future studies will focus on expanding the dataset, enhancing signal quality assessment methods, incorporating additional physiological features, and introducing subject-specific calibration techniques to improve both prediction accuracy and generalization performance. Furthermore, tracking individual blood pressure changes over time, rather than predicting absolute blood pressure values alone, may represent a promising research direction for wearable healthcare applications.

5. Conclusion

In this study, an end-to-end cuffless blood pressure estimation system based on synchronized ECG and PPG signals was developed and evaluated. The proposed framework integrates biosignal acquisition hardware, a signal processing pipeline, feature extraction algorithms, machine learning-based blood pressure prediction models, and validation procedures.

A total of 84 measurement samples were collected from 20 subjects and analyzed. Biosignal-derived features, including PAT and PTT, together with demographic information, were utilized to estimate systolic and diastolic blood pressure.

Among the three evaluated regression models—SVR, Random Forest, and Hybrid LASSO-XGBoost—the Hybrid LASSO-XGBoost model demonstrated the best overall performance, achieving mean absolute errors of 6.90 mmHg for SBP and 6.74 mmHg for DBP. These results demonstrate the feasibility of ECG/PPG-based non-invasive blood pressure estimation.

The study also revealed both the potential and limitations of PAT/PTT-based blood pressure estimation. In particular, challenges related to subject-independent generalization, dataset scalability, feature enrichment, and personalized calibration remain to be addressed. Nevertheless, the findings provide an important technological foundation for the future development of wearable continuous blood pressure monitoring systems.

Although the current system does not yet fully satisfy clinical-grade standards such as ISO 81060-2, this work is significant in that it successfully integrates the entire development process, encompassing hardware design, signal processing, machine learning modeling, and clinical validation. With larger datasets and more advanced algorithms, the proposed framework has the potential to evolve into a more accurate and reliable cuffless blood pressure estimation system.

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Author Contributions

Eunseol Lee (Hardware Team Lead) conceived and supervised the hardware architecture for simultaneous ECG and PPG acquisition. Designed and integrated the AD8232 ECG module, MAX30102 PPG sensor, and STM32 microcontroller-based acquisition platform, and led sensor interfacing, power management, hardware–software integration, and system validation.

Yujin Kim, Juha Choi, Seongyeong Lee, Yebin Moon, Siyoung Park, and Chanju Kim implemented ECG and PPG data acquisition using ADC and I²C communication interfaces. Developed the timer-based sampling framework, optimized ECG electrode placement and PPG sensor attachment conditions, analyzed noise sources and contact-related issues, and contributed to signal quality improvement through lead-off detection verification and sensor performance evaluation.

Seonghyeon Jeong (Signal Processing Team Lead) led the signal processing team and established the theoretical foundation for ECG/PPG analysis. Defined the signal processing pipeline architecture, coordinated task decomposition and parallel development, and facilitated inter-team communication regarding interface design and system integration.

Seongyun Jo developed the time synchronization and data standardization framework for ECG and PPG signals. Proposed a unified data structure containing timestamps and sample indices, implemented segmentation and sampling-period correction algorithms, coordinated data collection formats with the hardware team, and standardized data handoff procedures for the machine learning pipeline.

Jieun Park investigated criteria for identifying low-quality physiological signals and contributed to the initial design of ECG filtering methods using band-pass filtering techniques. Participated in waveform inspection, pipeline debugging, and data quality verification.

Inwoo Seong developed the ECG R-peak detection algorithm based on the Pan–Tompkins methodology. Implemented derivative, squaring, moving-window integration, dynamic thresholding, and local maximum refinement procedures. Additionally, developed automated visualization and CSV export tools for detection results.

Chanwoo Lee investigated PPG foot point detection methods and implemented algorithms for extracting foot points, PAT, PTT, and RR interval features. Contributed to the development of the physiological signal analysis pipeline and automated feature generation framework.

Siwoo Kim (Machine Learning Team Lead) led the machine learning team and managed project progress and inter-team collaboration. Designed the machine learning framework for blood pressure estimation and developed the Hybrid LASSO-XGBoost model. Implemented data augmentation techniques (Uniform, Gaussian, Dynamic Gaussian, and Mixed augmentation), virtual dataset generation, hyperparameter optimization, automated validation pipelines, and error distribution visualization tools.

Subin Han developed Support Vector Regression (SVR)-based blood pressure prediction models and proposed a prediction framework incorporating the physiological relationship between SBP and DBP. Designed the training pipeline, implemented hyperparameter optimization procedures, performed model validation using synthetic and real-world datasets, and contributed to code refactoring and performance analysis.

Sumin Kim developed Random Forest-based blood pressure prediction models and investigated hyperparameter optimization strategies. Implemented Optuna-based optimization and Leave-One-Group-Out (LOGO) cross-validation, developed feature importance visualization tools, and designed separate optimization frameworks for SBP and DBP prediction.

Yunji Oh improved the Random Forest framework by expanding hyperparameter search spaces and implementing Bayesian optimization using Optuna. Developed IQR-based outlier removal methods and optimized preprocessing parameters to balance noise reduction and data retention in small-scale datasets.

Chanmi Kim developed SVR models with RBF kernels and implemented hyperparameter optimization using RandomizedSearchCV. Established model evaluation metrics, including MAE, RMSE, R^2, AAMI criteria, and BHS grading. Investigated multiple public cuffless blood pressure datasets, identified and corrected data leakage and preprocessing issues, and performed extensive model validation.

Minje Kim (Validation Team Lead) led the validation team and established physiological validation protocols and model evaluation criteria. Investigated blood pressure estimation theory and clinical standards, supervised data quality assessment, and coordinated interpretation of experimental results.

Jiyeon Yu, Jiyun Shin, Minseo Kim, and Dongmin Lee developed measurement protocols and subject management guidelines. Designed procedures to control physiological confounding factors such as caffeine intake, physical activity, and posture. Established signal quality index (SQI) criteria and data rejection protocols and evaluated model performance according to AAMI/ESH/ISO 81060-2 standards

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